Magnetic Properties and Normal and Reversed Natural Magnetization in the Mull Lavas.

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Summary

The previous paper by Dr J. Ade-Hall has shown a petrological distinction between normal and reversed lava specimens from the Isle of Mull, Scotland. An investigation of several magnetic properties shows, surprisingly, that only the behaviour of the natural moments shows any distinction between normal and reversed specimens. The implication is that microscopic investigation is a necessary tool for the proper interpretation of many palaeomagnetic results, since it can reveal variations of properties unnoticed in a purely magnetic investigation. In particular, the problem of self-reversal versus field reversal demands petrological as well as magnetic investigation.

1. Introduction

In 1953, S. K. Dutt (1955) collected and measured the natural magnetizations of a large number of Tertiary lava samples from the Isle of Mull, off Scotland. Of 83 flows recorded, 48 were sampled, and 10 or 12 independently oriented samples were taken from each flow. Most of the flows were olivine flood basalts. Dutt found many samples with intermediate directions of natural magnetization, but there were also two definite groups of directions, one group associated mainly with the upper flows and clustered fairly close to the present dipole field direction at Mull, and the other group associated with the lower flows and magnetized in the reverse direction. Attempts to use alternating fields and thermal tests up to 200°C, did not reveal fully why the intermediate directions existed, and Dutt tentatively concluded that they indicated that the Earth's magnetic field has reversed its polarity a few times during the deposition of these lavas. However, single lavas of predominantly one polarity were occasionally found interleaved with oppositely polarized lavas. Eleven individual lavas even contained both normally and reversely magnetized specimens. This was very hard to explain by the hypothesis of field reversal during extrusion.

In a cursory petrological survey of thin sections, Dutt found no petrological difference between normal and reversed specimens. In 1961, Dr J. Ade-Hall examined more thin sections of these samples and found certain petrological differences between normal and reversed specimens from flow number 100. A very concentrated further examination both of thin sections and of polished surfaces has been reported in the preceding paper, where Ade-Hall shows that distinct

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petrological differences between normal and reversed specimens exist both in the opaque and transparent minerals. The obvious next step was to examine the magnetic properties of the rocks in considerable detail, in order to try to discover magnetic differences corresponding to the petrological ones. The magnetic examination is the subject of this paper.

Section 2 is a re-presentation in condensed form of some of Dutt's measurements on the original specimens. We thank Dr Dutt for his permission to use this data. Section 3 contains the effects of alternating fields and heatings on the natural magnetizations. Section 4 presents the results of standard magnetic measurements —saturation moment, saturation remanence, remanent coercive force, Curie point and initial susceptibility. Section 5 reports a detailed examination of ten samples from flow 100—six normal and four reversed. Section 6 contains a general discussion and an attempted interpretation of the data as a whole.

2. The natural magnetization of the Mull Lavas

The Mull lavas have been numbered from 44 to 126. Only those above flow 58 are positively known to descrease in age as their number increases, and following Dr Ade-Hall I have in general considered only flows between 58 and 126. However I have plotted all the natural magnetization data from 44 to 126 because it was already available, and because the flows 44 to 57 were known to lie very low even if their absolute chronological order was unknown. Dutt's measurements have been divided into five groups corresponding to those flows lying between numbers 44–65, 67–79, 81–100, 103, 104 and 105 (three special cases), 106–126.

Figure 1a shows stereoscopic diagrams, for each of these groups, of Dutt's specimen directions. In order to separate normal and reversed groups distinctly, directions are plotted on a diagram with the observer looking from the east, rather than from above as is usual. No distinction was found between eastward and westward directions, and none is made on the diagrams of Figure 1a. The overall mean axis of magnetization (353°, 67·5° down) which Dutt deduced from his most consistent lava groups, is represented by the black line crossing the diagram. The short arrow is along the present dipole field. Reversed, normal and discordant directions have been defined as follows: circles of arbitrary radius 30° were drawn about each end of the mean axis as centre. Directions lying within the north and down circle were called normal (N); those within the south and up circle were called reversed (R); and the rest were called discordant (D). The histograms of Figures 1a and 1b show the distributions of intensities of natural magnetization for N, R and D specimens within each group of lavas and also for all of the lavas together. The key remarks concerning Figures 1a and 1b are:

(a) There is an obvious statistical shift from almost no normal specimens in the lowest group, to absolutely no reversed in the upper group.

(b) The distributions of natural intensities have significantly different shapes for normal and reversed rocks in each of the five groups containing both N and R, (Figure 1a) and also in the overall histogram for all of the flows (Figure 1b). Flows 103, 104 and 105 are exceptional with intensities ten times the average of the other normal specimens. This was checked and is not a numerical slip. It remains unexplained.

(c) Discordant intensities are not near to zero, but have wide spread similar to the reversed ones. Discordant directions have preponderantly southward declinations. We have deliberately concerned ourselves greatly with discordant specimens.

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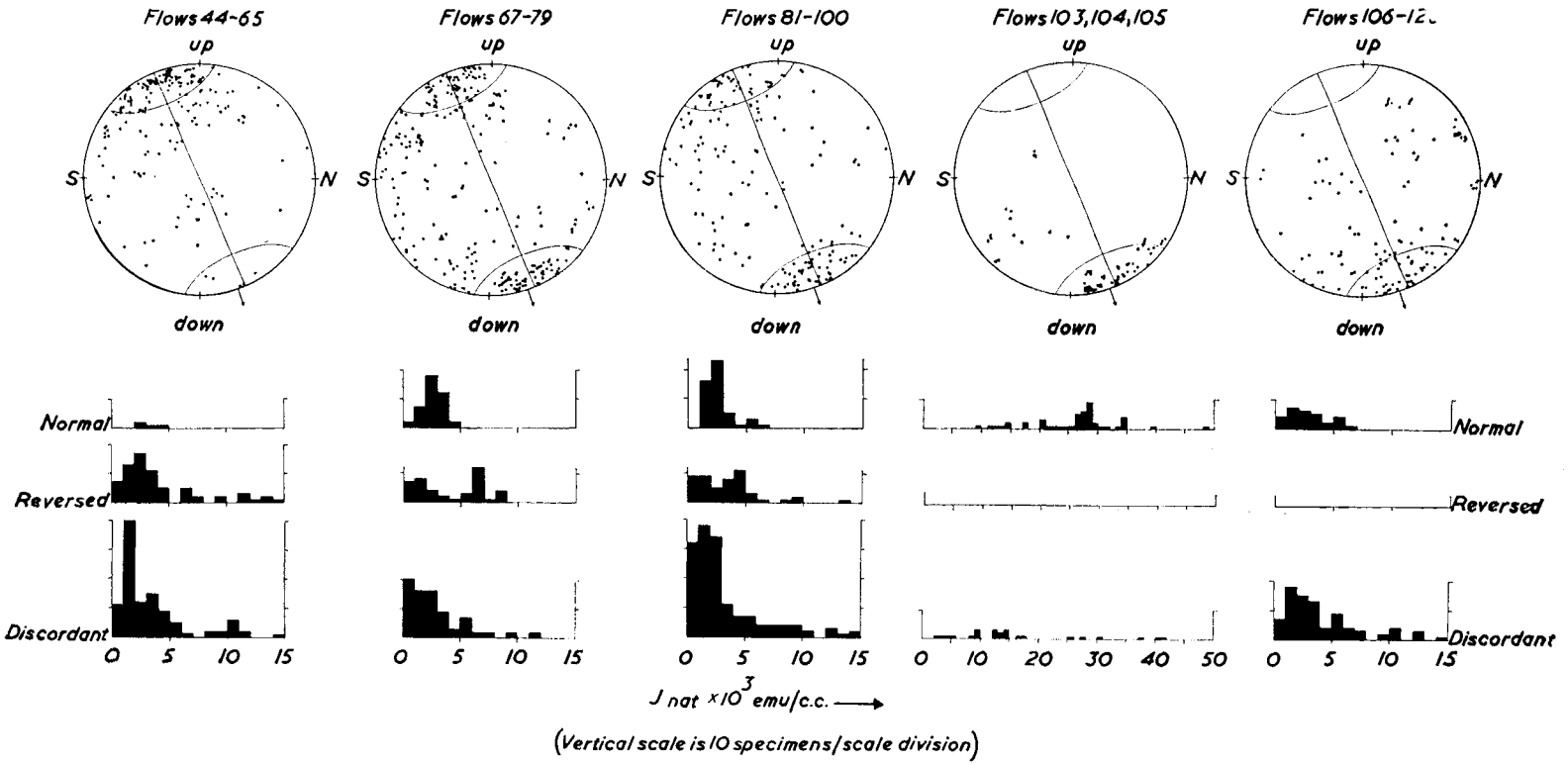


FIG. 1a.—Classified directions of magnetization and natural intensities for the Mull lavas.

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3. Alternating field and thermal demagnetization of the natural moments

The natural moments of seven virgin specimens (two normal, two reversed and three discordant) were progressively demagnetized in alternating fields up to 450 oersteds. The effects were not so smooth and clear that much faith could be put into them. I shall nevertheless describe the results. The discordant specimens tended to retain their original directions of magnetization with fairly regularly

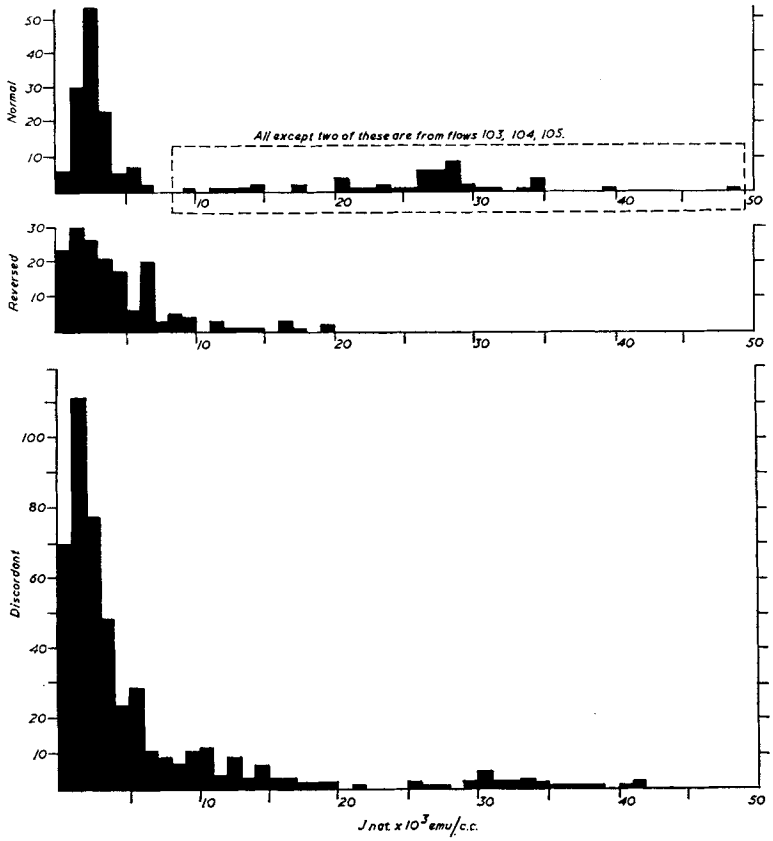


FIG. 1b.—Histograms of natural intensities for all Mull flows 44–126.

decreasing intensities. One reversed specimen became discordant, the other remained reversed. Both normal specimens became magnetized north and steeply up (discordant). The failure to produce a rational result may be due to insufficiently refined equipment. Dutt's attempts at A.C. demagnetization produced equally conflicting results, due to the crude state of the art in 1953 in Britain.

The natural moments of twelve virgin specimens (five normal, three reversed, and four discordant) were thermally demagnetized, while only the two components of the natural vector magnetization lying in the plane containing north–south and up–down axes were measured. This was because the already cubic specimens left by Dutt could not be adapted to three-dimensional measurement in our apparatus.

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(*a*) *The three reversed specimens*—(67/7I, 70/10I, and 76/6I*) retain their initial directions very closely up to their Curie points (570 to 590°C). Specimen 70/10I had an additional weakly magnetic constituent with Curie point about 650°C, with natural direction accurately parallel to that of the lower Curie point material.

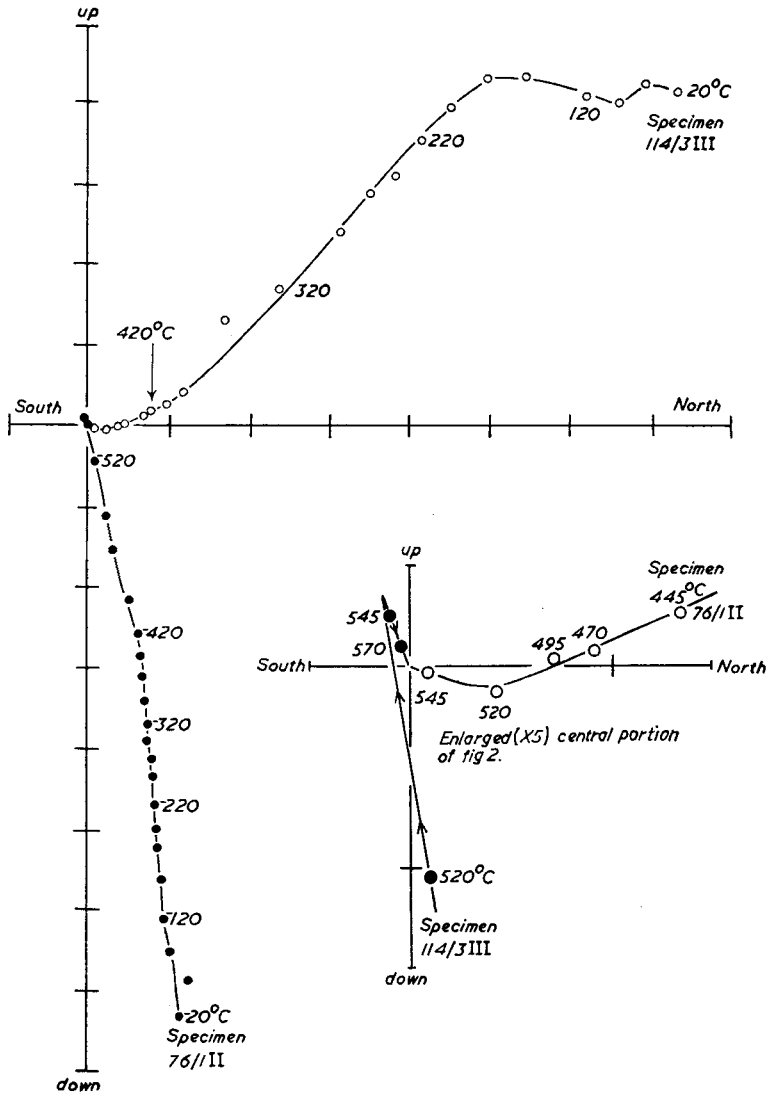


FIG. 2.—Change of natural magnetism during heating. Specimen 76/1 II reversed its sense 50° C below the Curie point. Specimen 114/3 III also changed direction near its Curie point. (25° C intervals with 45° C missing.)

(*b*) *Of the five normal specimens*—three of them (67/6I, 67/2II and 114/8I) retained their normal directions until they disappeared at 570–600°C. Another specimen (70/2I) did the same but disappeared at 530°C. Specimen 76/1II (Figure 2) inverted to become reversely magnetized between 520 and 570°C.

* 67/7I means flow 67, seventh oriented sample from flow 67, first cube cut from seventh sample.

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(*c*) *Of the four discordant specimens*—61/3I became definitely reversed at temperatures above 320°C. Specimen 98/9 became normal above 195°C; its natural moment disappeared just above 320° C, although two separate virgin pieces in 1000 oersteds had Curie points of about 600°C. Specimen 114/3III (Figure 2) had a presumed viscous magnetization which disappeared at 170° C. There was then a sudden change in the rate of decrease of intensity near 370° C, which is probably due to a magnetic material with this Curie point which is widespread in Dutt's rocks (section 4). Above 400° C, there was a gradual change of direction towards the normal sense. Finally, specimen 114/9II remained discordant (north and shallow down) at all temperatures.

Specimens 76/1II and 114/3III (Figure 2) may provide valuable clues to the nature of many of Dutt's rocks. They both show that the natural magnetic moment can either swing rapidly in direction, or reverse its sign, less than fifty centigrade degrees below the temperature of total disappearance. It is sometimes easy to miss these changes because the effect may be small and one tends to stop heating when the natural moment first becomes zero, on the assumption that it will not grow in the opposite sense at higher temperatures.

4. General magnetic properties

An attempt has been made to correlate normal, reversed and discordant magnetization with some general magnetic properties, namely:

(*a*) Saturation magnetization (J_{sat}) in 2 300 oersteds.
(*b*) Saturation remanence (J_{rem}) after exposure to a 10 000 oersted field.
(*c*) Remanent coercive force (H_{cr}), which is the back field that leaves the saturation remanence zero after its removal.
(*d*) Curie point, Tc, in a 2 300 oersted field.
(*e*) Weak field susceptibility, χ in 0·5 oersteds.

(*a*) *Saturation magnetization* J_{sat}—Specimen pieces weighing about 0·5 g were used. It first appeared that statistically J_{sat} was greater for normal than for reversed specimens (Figure 3a). However, this is not a causal correlation. Figure 3b shows that there is a tendency for higher-lying flows to have a greater J_{sat} than lower-lying ones. Since the upper flows are predominantly normal (and vice-versa), the correlation follows automatically. But within each single flow containing both normal and reversed specimens (and discordant), J_{sat} is quite constant and obviously independent of the natural polarity (flows 61, 67, 76, 98, 100 and 114 (boxed in Figure 3d)). There is therefore no causal correlation between J_{sat} and the polarity of the natural remanence.

(*b*) *Saturation remanence* J_{rem}—Figure 3b (like Figure 3a) shows a higher average J_{rem} for normal specimens, but this too is a misleading correlation for the same reason as in (a). A confirmation of this is that J_{rem} is roughly proportional to J_{sat}, so that again in higher flows one might expect higher J_{rem}, independent of the natural polarity.

(*c*) *Remanent coercive forces* H_{cr} (Figure 3c)—These lie mainly between 200 and 600 oersteds, and are not correlated with the natural polarity.

(*d*) *Curie Point Tc*—Two sets of heating curves were done to determine Curie points—one set on larger pieces (~0·5 g) from flow 100 in 2 300 oersteds (see section 5), and the other set on smaller pieces (~0·01 g) from various selected

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specimens in several hundred oersteds. In general two apparent Curie points were observed.

The higher Curie points of both sets fell mainly between 555 and 620°C (Figure 3e). The saturation moment was not altered much by heating in air to 600° C.

The lower Curie points, at about 400 to 450° C, always occurred in the larger specimens but only occurred randomly in the smaller ones, from which it seems that this low Curie point material is inhomogeneously distributed throughout the rocks. It may also be true that the transformation at 400–450° C is an irreversible

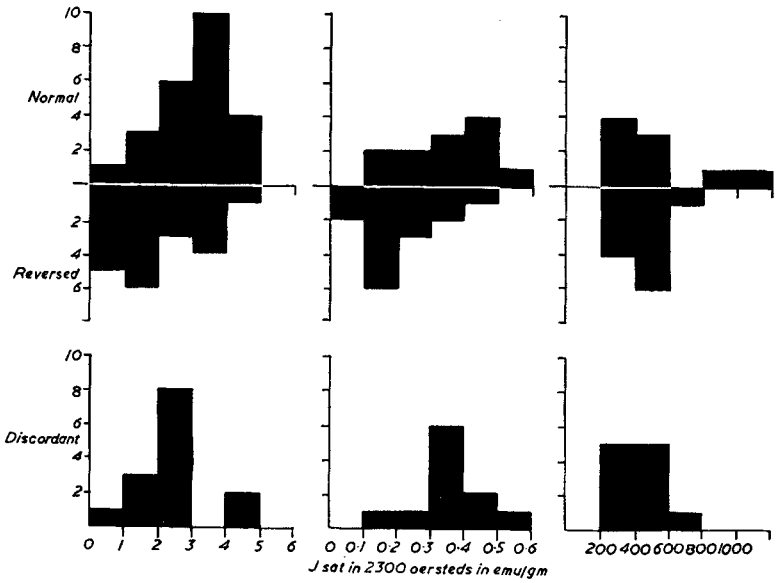


FIG. 3a.—J_{sat} in 2 300 oersteds, in e.m.u./g.
FIG. 3b.—J_{rem} after 10 000 oersteds, in e.m.u./g.
FIG. 3c.—Remanence coercive force, H_{cr}, in oersteds.

destruction of some magnetic material (e.g. maghemite to hematite), rather than a true Curie point, because it never reappears on cooling.

No Curie point or heating curve data showed any correlation with the natural polarity. The lower "Curie point" material certainly is not connected directly with the normal and reversed state, because some rock specimens switched direction or went from normal to reversed well *above* 450° C.

(*e*) *Weak field susceptibility*, χ—Figure 3f shows the distributions of weak field susceptibility χ in 0·5 oersteds for normal, reversed and discordant specimens, as determined by Dutt. There is here a significant difference between normal and reversed specimens, although the following section 5 contains evidence to show that this difference too may be coincidental rather than being causally linked with the natural polarity.

5. Investigation of flow 100

Flow 100 was the first instance found in the Mull Lavas of the petrological differences between normal and reversed specimens. This flow is quite ideal for a specialized investigation because, of ten oriented samples collected, six had

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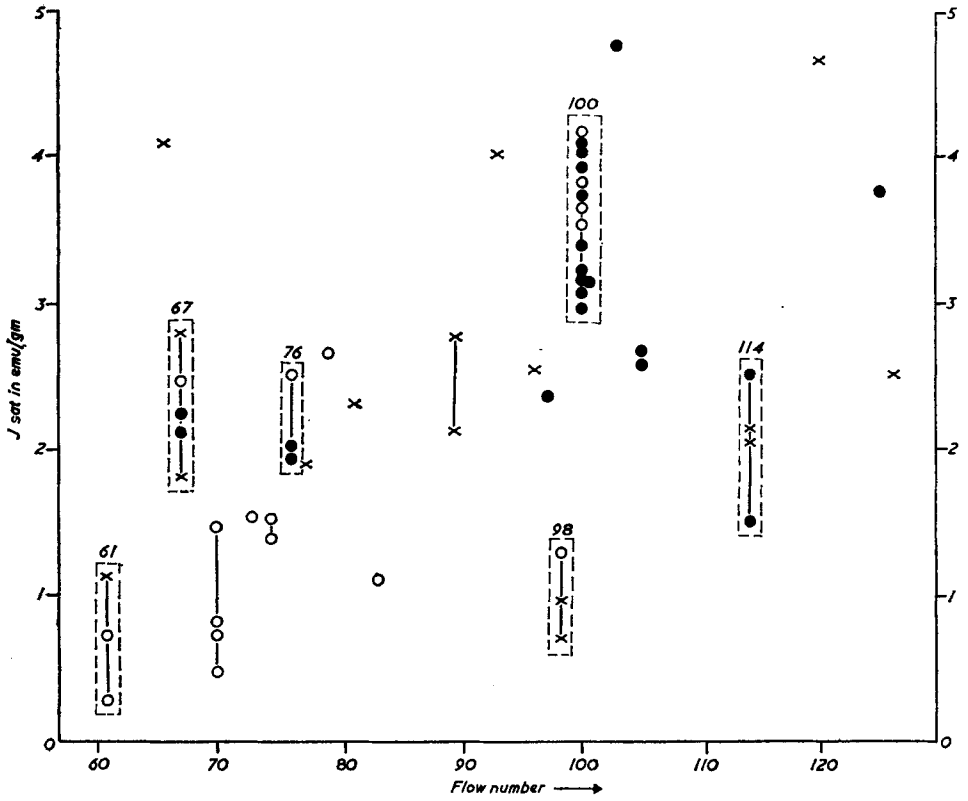


FIG. 3d.—J_{sat} for specimens from selected flows. Within each flow, J_{sat} does not depend on the natural polarity (boxed-in values).

● Normal
× Discordant
○ Reversed

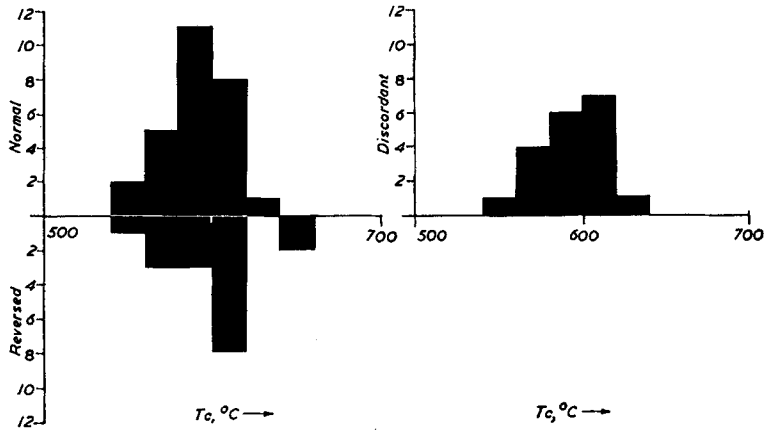


FIG. 3e.—Histograms of Curie points for normal, reversed, and discordant specimens.

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normal natural directions of magnetization and four had reversed. The petrological differences between normal and reversed specimens from flow 100 are immediately obvious in thin sections and polished surfaces, although the basic rock type is the same for all samples, showing that they can indeed have come from the same flow, as Dutt has recorded.

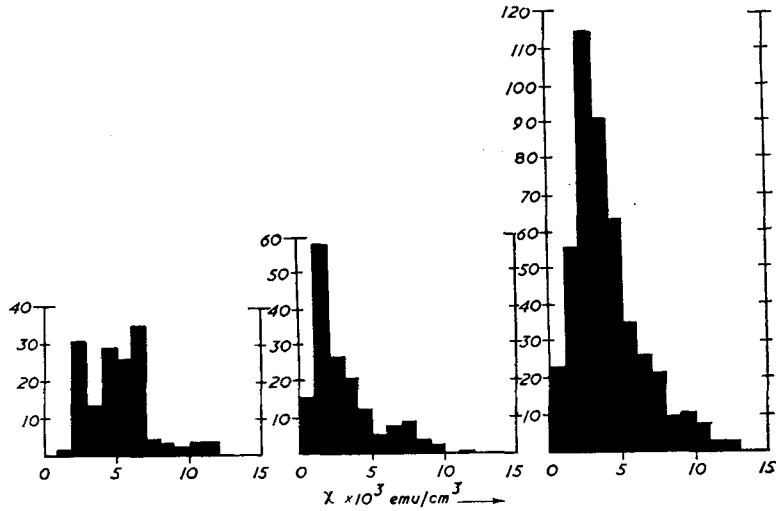


FIG. 3f.—Weak field susceptibility χ, for normal, reversed and discordant specimens.

Table 1

Curie points in 2 400 oersteds for both normal and reversed specimens from the single flow 100.

Specimen	Natural polarity	Curie point in 2 400 oersteds
100/1I	Normal	590
100/2I	,,	610
100/3II	,,	600
100/4II	,,	590
100/5	,,	—
100/6I	,,	600
100/7	Reversed	600
100/8I	,,	620
100/9	,,	600
100/10II	,,	610

The magnetic characteristics of these samples are shown in Figures 4a to 4d. Figure 4a shows tight grouping of normal natural intensities, but a spread of reversed intensities, verifying the same general relationship for the lavas as a whole (Figure 1b). On the other hand, the saturation moments are identical in normal and reversed pieces (Figure 4b), which is also generally true for any given flow (section 4a).

There is no significant difference between χ for normal and reversed specimens in flow 100 (Figure 4c), and this led us to believe that the general differences in Figure 3f are probably unconnected with natural polarity.

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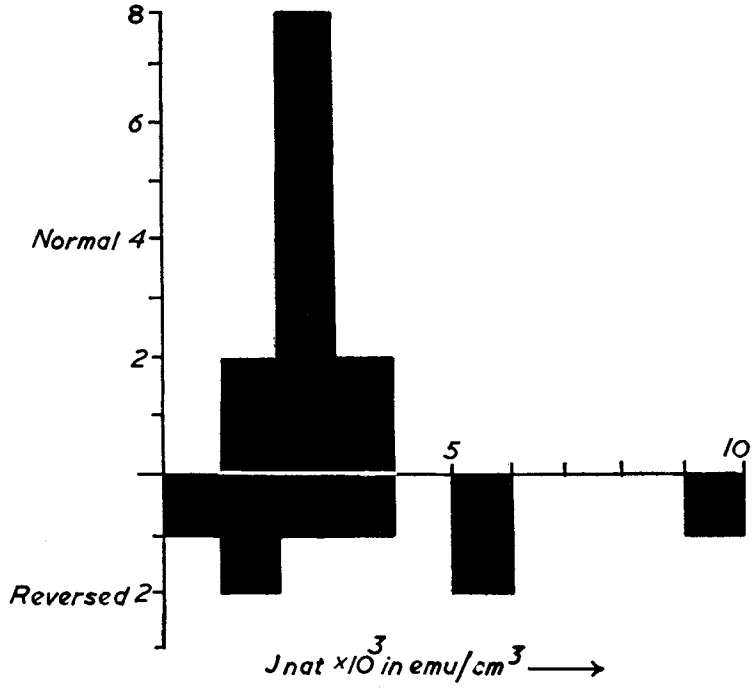


FIG. 4a.—Distribution of normal and reversed natural intensities in flow 100. Normal intensities are grouped, reversed are scattered.

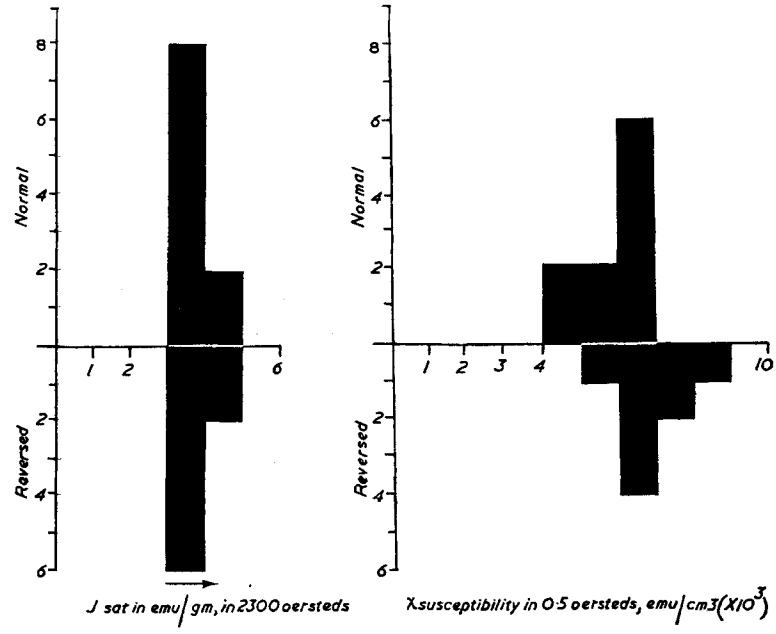


FIG. 4b.—Showing no significant differences in J_{sat} between normal and reversed specimens from flow 100.

FIG. 4c.—Showing no significant differences in χ between normal and reversed specimens from flow 100.

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The higher Curie points (in 2 300 oersteds) of ten flow 100 specimens were equal to 600°C , within ± 10° C scatter (Table I). Each specimen also had a lower transition point at 400–450° C, on heating only.

The natural moments of the six normal and four reversed specimens were investigated with increasing temperature. Only the magnetic components in the

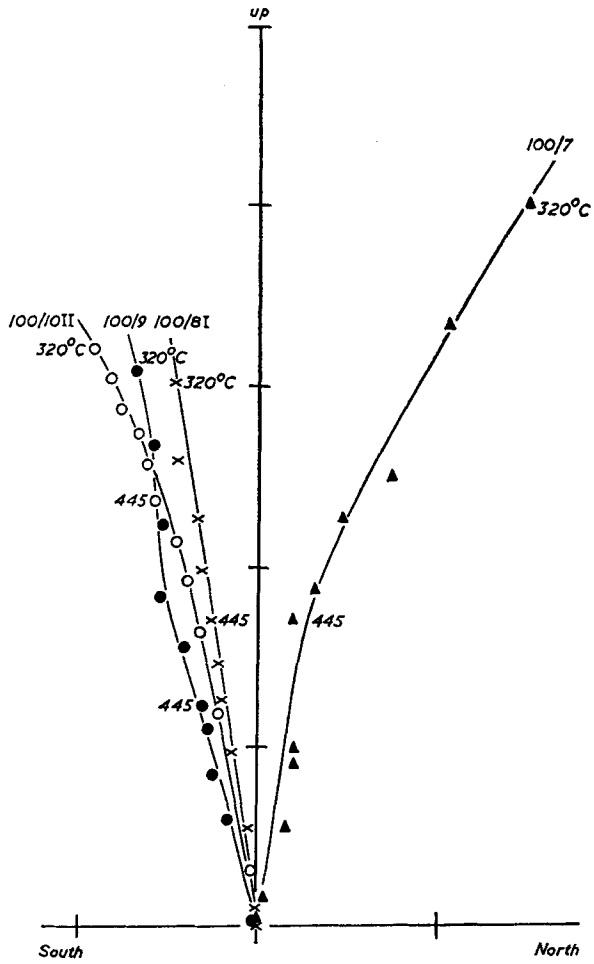


FIG. 4d.—Change of natural vector magnetization during heating of four reversed specimens from flow 100. For clarity, only points above 320° C are shown, although there was also little change of direction at lower temperatures.

north–south and up–down plane were measured, since the cubic specimens were ill-adapted to three-dimensional measurement in our apparatus. There are three general observations about the results:

(*a*) The four reversed specimens (Figure 4d) remained reversed, changed direction little, and decreased smoothly in intensity up to their Curie points near 570 to 600° C.

(*b*) The normal specimens behaved on the contrary in a very anomalous manner. To avoid complexity, Figure 4e shows their behaviour only above 320° C. All but one of them (100/3 II) tended to become reversed, some

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of them less than 50° C below their Curie points (about 570–600° C). The only one which remained indisputably normal (100/3 II) disappeared at the anomalously low temperature of 495° C (although readings were taken up to 670° C to ensure that the moment remained zero). The Curie point of (100/3 II) in 2300 oersteds was nevertheless about 600° C.

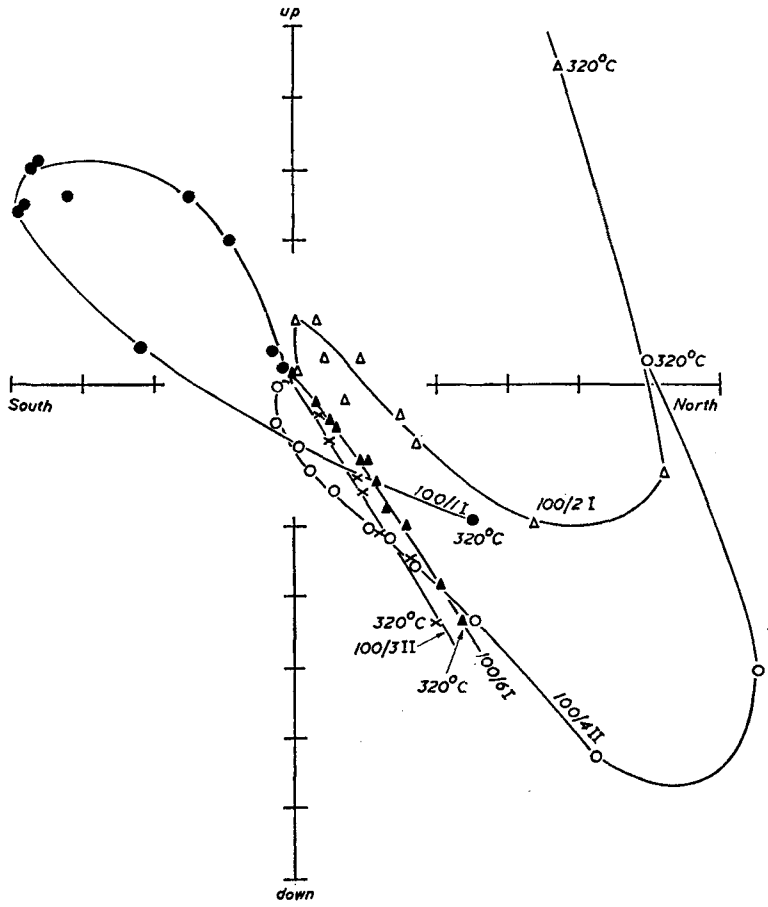


FIG. 4e.—Change of natural vector magnetization during heating of five normal specimens from flow 100. All were "normal" at room temperature. These plots begin at 320° C so as to show the high temperature detail. The direction changes are multiple and complex. All but one tended towards the reversed state at highest temperatures (see text).

The reversed specimens behaved in no way anomalously, while all the normal ones did behave anomalously. This is not meant to imply that the original moment must necessarily have been a reversed one, for there is no guarantee of originality for *any* of the moments observed.

6. Summary and conclusions

The central facts, extracted from these diverse results and from Dutt's thesis are:

(*a*) Eleven single lavas each contained both normal and reversed samples.

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(*b*) In a strong field, of a few hundred oersteds or more, no relevant difference of any kind was found between normal, reversed and discordant specimens.

(*c*) The one weak field property, the susceptibility χ in 0·5 oersteds showed, between normal and reversed rocks, a statistical difference which is unlikely to be real, because it is not apparent internally in flow 100 (section 5).

(*d*) The only obvious magnetic distinctions between normal and reversed rocks lay in the behaviour of the natural moment. The normal natural intensity distribution was peaked more sharply than the reversed. The discordant rocks did *not* have relatively weak natural intensities.

During thermal demagnetization, the normal magnetizations often varied anomalously and sometimes changed sign at high temperatures, while none of the reversed specimens behaved anomalously.

(*e*) The natural moments of three specimens disappeared at 530° C, 495° C and 300° C, although their Curie points in 2300 oersteds were 600° C. All three were normal specimens.

Any conclusions to be drawn from these facts must be considered in the light of the geological history of Mull. The area of these lavas contains many Tertiary dykes. It is also on the edge of a zone of pneumatolysis centred on the two volcanic caldera about two miles away. The flows numbered from 100 up to 114 are also within a quarter of a mile of a large granophyre intrusion (see Map 1 in Ade-Hall's paper).

It must also be kept in mind that the normal and reversed specimens differ greatly and certainly significantly in their petrology (previous paper). By reflection microscopy, the normal specimens have homogeneous opaque titanomagnetite grains while the reversed ones have exsolved ilmenite lamellae in these grains.

There are then a great many experimental results to consider in discussing the significance of the natural polarization of the Mull lavas. I have chosen first to discuss the natural magnetizations in isolation, first the normal and reversed specimens, then the discordant specimens.

The *normal and reversed* specimens pose the old problem of self-reversal versus field reversal. One can easily construct *ad hoc* hypotheses which support either contention, by inventing local and/or general reheatings by the dykes, sills, and caldera known to be present, and also by postulating field reversals at various stages of the heating history of the rocks. The field relationships are so imprecisely known as to make fruitless such hypotheses. On the other hand, it is quite difficult to find even one good explanation of the nature of the discordant specimens.

Discordant specimens. A major difficulty of interpretation is to explain the directions and magnitudes of the natural moments of the discordant specimens. Figure 1a shows that they are directed preponderantly south rather than north. Of 423 discordant specimens, 279 (66 per cent) are in the southern half, and 144 (34 per cent) in the northern. Only the highest set of flows violates this rule. The natural intensities have a very similar distribution to that of the reversed rocks, but are more spread out to low and high values than the normal ones (Figure 1a) (excepting flows 103, 4 and 5 as before).

The discordant specimens are unlikely to have been remagnetized subsequent to collection because their directions are not random, and also because most of them remained unchanged over a storage period of eight years in the Earth's field.

There remain several possibilities:

(*a*) that the discordant rocks were reversely magnetized but partially unstable

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in the Earth's present (normal) field over a long time. This cannot be so, because some of the discordant specimens became *normal* on heating (section 3), and also because one would expect simply unstable specimens to have lower average intensities (or at least much different) intensities than either fully normal or fully reversed specimens, since their magnetization would be a superposition of two nearly antiparallel vectors.

(*b*) that the discordant rocks represent an intermediate stage of self-reversal. The petrological differences between normal and reversed rocks suggest this; but intermediate directions ought certainly then to have lower average intensities which is not so. They ought also to be randomly oriented, being the small differences between large vectors (summed over a specimen). These conclusions do not fit the facts, and it seems that here is an argument against the self-reversal hypothesis for normal and reversed specimens.

(*c*) that the discordant specimens are due to lightning strokes. This hypothesis can be reconciled neither with the strong tendency to southerly declination in the discordant rocks, nor with the similar intensity distributions for reversed and discordant specimens. The coincidence would be too great considering the random nature of lightning. Dutt found a few other specimens with intensities up to 1 000 times the ones discussed here. These are much more likely to be due to lightning although not necessarily so. The discordant rocks mentioned in this paper are certainly not the "tail" at the lower end of this lightning induced distribution. Lightning cannot explain these discordant rocks.

(*d*) that the discordant rocks are in some way recording the Earth's field during periods of reversal. From the evidence available, this could explain the non-random distributions of discordant directions (van Zijl *et al* 1962, Momose 1958, Brynjolfsson 1957, Khramov 1958). But the discordant specimens all (except one) swing during heating to either normal or reversed directions. To explain this requires some extra hypothesis such as acquisition of partial thermo-remanences by slow cooling during switching of the earth's field polarity. This is too *ad hoc* to be generally acceptable.

The complete evidence will now be considered in relation to the natural polarity. The essence of this discussion is that, surprisingly, the general magnetic properties bear no relation to the petrological observations on opaque grains or to the properties of the natural magnetizations. The complete evidence falls naturally into the three categories—general magnetic properties, petrological properties, natural magnetization properties—and these three will be discussed in pairs.

Conflict between the natural remanence and the general magnetic properties

The normal natural remanences are distinct from the reversed ones first in the different distributions of natural intensities, second in the complex behaviour of some of the normal remanences during heating as contrasted with the simple behaviour of the reversed remanences. The general magnetic properties bear no relation to these differences. For example, the tightly grouped normal intensities and the more widely spread reversed intensities (Figure 1a) do not correspond to the grouping of saturation moments in the histograms of Figure 3. It then seems that the strength of the natural moment bears little relation to the *amount* of magnetic material as reflected in the saturation moment. The other general magnetic properties, with the just possible exception of weak field susceptibility χ

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show no more correspondence with the natural polarity than does the saturation remanence. Even χ does not show a variation from normal to reversed specimens when critically tested on the single flow 100.

In addition, the three normal specimens (70/21, 98/9 and 100/3 II had the usual high field Curie points of 600° C while the natural moments disappeared at 530° C, 320° C and 495° C respectively.

These facts seem to imply one of the following two statements:
either:

(*a*) the normal and reversed magnetic states are due to the different thermal and magnetic *histories* of the specimens (such as partial thermoremanences and switching of the Earth's field), so that the natural state is not fully dependent on only the intrinsic magnetic properties, or

(*b*) the natural moment does not in fact reside in these magnetic minerals which predominate in strong fields (i.e. which exhibit the general properties). This is not without precedent. Both Uyeda (1958) and Carmichael (1961) have discovered the natural remanence in certain rocks to reside in a few per cent or less of the total potentially magnetic material, the rest being in a demagnetized state, but capable of strong magnetization in large fields.

Neither of these possibilities decides in favour of field reversal or self-reversal, although the first might be taken as an *indication* of field reversal.

Comparison of the petrological characterstics wth the state of natural magnetization

Ade-Hall has done this comparison very thoroughly in the preceding paper. There is no doubt about the strong correlation of exsolved ilmenite in the opaque grains with reversed natural magnetization, and the equally strong correlation of unexsolved homogeneous opaque grains with normal natural magnetization. There are other similar petrological correlations which we think at the moment to be of a secondary nature. It seems reasonable to assume that either the process of exsolution has itself been a cause of the present natural polarization, or else the exsolution and the state of natural polarization are both the effects of a common cause, such as local reheating. For example, field reversal could have taken place during local reheatings which caused exsolution of ilmenite and hence a sort of "exsolution remagnetization" of the rock along the ambient reversed field (Bewersdorff 1961, Verhoogen 1962); or, self-reversal in a normal field could have been a result of the process of exsolution caused by local reheating. Bewersdorff has effected "exsolution remagnetization" of basalts without any self reversal coming to light, which suggests that the two states of the Mull lavas are due to original and later magnetizations in fields of opposite polarities. This remains to be proved by more extensive experiments.

Conflict between the bulk magnetic properties and the petrology of the opaque grains

Reversed specimens have 20 to 30 per cent of exsolved ilmenite in opaque grains; normal ones have (optically) homogeneous grains. This difference is not in the slightest reflected in the magnetic properties J_{sat}, J_{rem}, T_c and H_{cr}. Equally difficult to explain are the well defined Curie points near to that of magnetite in *all* the N and R samples. With so much titanium present, one would expect at least some lower Curie points. A few of the natural moments disappear at lower

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temperatures and this prompts the question: are the opaque grains the real seat of the natural magnetic properties?

The chief lesson we have drawn from these magnetic investigations is that the general magnetic properties of lavas have failed to distinguish between normal and reversed specimens which are manifestly and drastically different under the microscope. It seems that we can never again trust magnetic tests alone to this task of discrimination. Any future attempts must include petrological examination, especially of polished surfaces. This was pointed out to us long ago (Nicholls 1956, Balsley & Buddington 1957), but it took the real collaboration of petrologist and physicist to drive the fact home by experiment.

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